Fraud Detection in Financial Transactions

Data Science Case - Artefact

Leonardo Correia

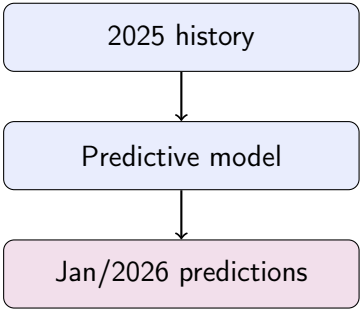
July 2026

Case objective

Business problem

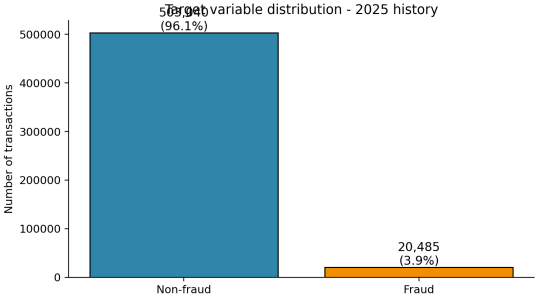
Automate the identification of potentially fraudulent transactions, reducing dependence on manual review and increasing process scalability.

- Minimize undetected fraud cases.
- Avoid unnecessary blocks of legitimate customers.
- Deliver classifications for the January/2026 dataset.



Data overview

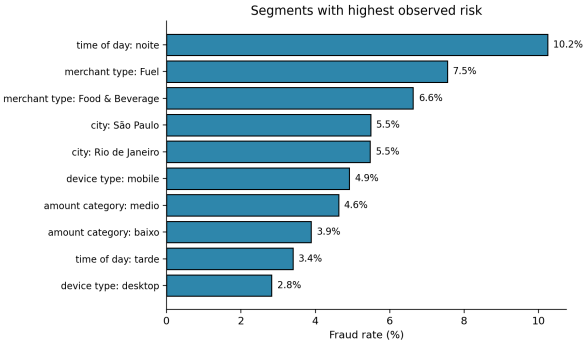
- Historical dataset: **523,525 transactions** from 2025.
- Target variable: `flag_fraude`.
- No missing values in the original columns.
- January/2026: **44,451 transactions** to classify.



Statistical challenge

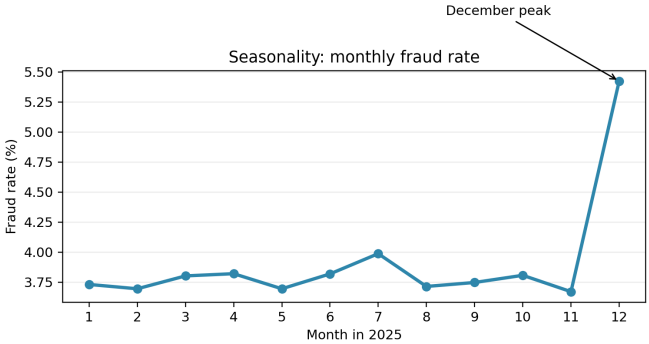
Only **3.91%** of historical transactions are fraud. Both the metric and the threshold must reflect this strong class imbalance.

Main exploratory patterns



- Higher risk in transactions during the **night** period.
- **Fuel** and **Food & Beverage** merchants concentrate higher fraud rates.
- **Mobile** shows higher risk than desktop and tablet.
- Sao Paulo and Rio de Janeiro appear among the cities with the highest rates.

Temporal pattern

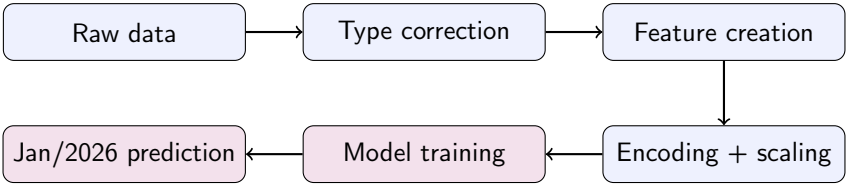


Insight

The monthly rate is relatively stable during the year, with a relevant increase in December.

- The month was used as a temporal feature.
- A recent-history variable was also created: frauds in the last 30 days.

Feature engineering and preprocessing



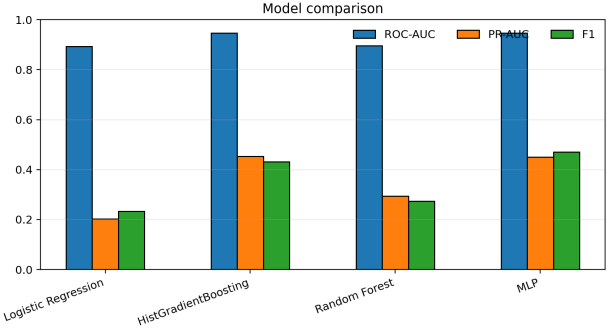
Categorical variables

`tipo_estabelecimento`, `cidade`, `device_type`, `turno`, `valor_categoria`

Numerical variables

`month`, `fraudes_ultimos_30d`

Evaluated models



Model	PR-AUC	Recall	F1
Logistic	0.203	0.822	0.233
HistGradientBoosting	0.453	0.737	0.432
Random Forest	0.293	0.209	0.274
MLP	0.450	0.633	0.471

- High ROC-AUC alone is not enough for imbalanced data.
- PR-AUC and the precision/recall balance were prioritized.

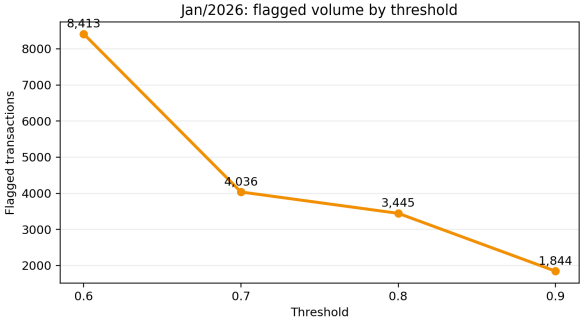
Selected model: HistGradientBoosting

Why this model?

- Best PR-AUC among the traditional models tested.
- Supports class weights through `sample_weight`.
- Good recall without depending on a very low threshold.
- More operationally stable than the MLP for this case.

Metric	Value
Threshold	0.70
ROC-AUC	0.946
PR-AUC	0.453
Precision	0.305
Recall	0.737
F1-score	0.432

Threshold as a business decision

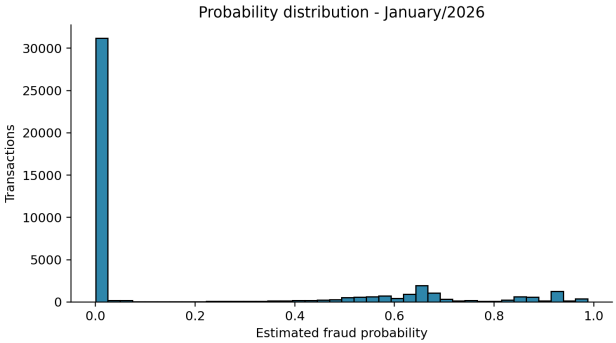


Interpretation

- Lower threshold: more flagged transactions and broader risk coverage.
- Higher threshold: fewer alerts and greater operational selectivity.

Threshold	Jan/2026 alerts
0.60	8,413
0.70	4,036
0.80	3,445
0.90	1,844

Final result for January/2026



Delivered file

`transacoes_jan2026_predictions.csv`

- `fraud_probability`: estimated probability.
- `fraud_prediction`: final class.
- Delivery threshold: **0.90**.
- Total flagged: **1,844 transactions**.

Production recommendations

- Define the threshold based on the cost of false positives versus false negatives.
- Monitor monthly drift in variables such as city, time of day, device, and merchant type.
- Recalibrate probabilities if the actual fraud rate changes in 2026.
- Track precision, recall, PR-AUC, and alert volume through a continuous validation routine.

Conclusion

Summary

A fraud classification solution was developed with exploratory analysis, feature engineering, model comparison, and prediction generation for January/2026.

- The problem is strongly imbalanced: PR-AUC and recall are central metrics.
- HistGradientBoosting was selected for its balance between performance and operational feasibility.
- The solution delivers both probability and final classification, allowing threshold adjustment according to the company's strategy.